Vigneshwaran Krishnamurthy

Specially Appointed Assistant Professor

Astronomical Institute, Graduate School of Science, Tohoku University · Sendai, Japan

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Academic Appointments

Specially Appointed Assistant Professor June 2026–present
Astronomical Institute, Graduate School of Science, Tohoku University; Sendai, Japan
Research on exoplanet atmospheres, planetary-system architecture, and high-resolution spectroscopy.

Visiting Researcher January–April 2026
Astrobiology Center, NINS; Tokyo, Japan
Instrumentation development and observations with a new high-resolution spectrograph.

Postdoctoral Researcher Nov 2022–April 2026
Trottier Space Institute, McGill University; Montreal, Canada
Atmospheric characterization of transiting exoplanets using ground- and space-based telescopes.

Project Research Staff Oct 2021–Sept 2022
Exoplanet Search Project Office, Astrobiology Center, NINS; Tokyo, Japan
Atmospheric characterization of young transiting planets using the InfraRed Doppler spectrograph on the Subaru Telescope.

Professional Development

Mini-MBA: Executive Development Course October 2025
Desautels Faculty of Management, McGill University
Intensive executive training in leadership, strategy, and innovation for research and technology professionals.

International Diploma in Astrobiology March 2016
Indian Astrobiology Research Centre; Mumbai, India
IARC International Diploma Program; Excellence Grade: A.

Education

Doctoral Research in Exoplanet Science September 2021
Tokyo Institute of Technology; Tokyo, Japan
Thesis: *Atmospheric Evolution of Low-Mass-Star Planets: A Tale of Evolution*.
Recipient of the Monbukagakusho (MEXT) Scholarship from the Government of Japan.

Master's Degree in Aerospace Engineering May 2017
Indian Institute of Technology Kanpur, India; GPA: 9.0/10.0
Recipient of the President Dr. Shankar Dayal Sharma Gold Medal.

Bachelor's Degree in Aerospace Engineering June 2015
Amrita Vishwa Vidyapeetham; Coimbatore, India; GPA: 8.58/10.0

Honors and Achievements

- **2026** **Canadian Space Agency JWST Cycle 4 Grant**, Montreal, Canada. Grant to study a multi-planet system with JWST.
- **2025** **Arena FIDE Master Title (AFM) in Chess**, Montreal, Canada. Achieved the Arena FIDE Master title in early 2025.
- **2024** **Private Pilot Training**, Montreal, Canada.
- **2023** **Canadian Transnational U1300 Chess Champion**, Montreal, Canada.
- **2020** Graduate Student Assistant–Program Developer (**GSA-D**), Tokyo Tech, Japan.
- **2018** **Recipient of the Monbukagakusho (MEXT) Scholarship**, Government of Japan, Tokyo Institute of Technology, Japan.
- **2017** **President Shankar Dayal Sharma Gold Medal**, Indian Institute of Technology Kanpur, India. Highest honor for graduate studies at IIT Kanpur.
- **2016** Published the novel *The Forest Whisperings* with Partridge Publishing, Coimbatore, India.
- **2016** **Best Paper Award for Motivational Research**, ICMME 2016, Shanghai, China.
- **2015** **Government of India Assistantship for Graduate Studies**, Indian Institute of Technology Kanpur, India.
- **2009** **Honorable Mention**, NASA Space Settlement Contest, NASA Ames Center, USA.

Publications

Primary-author publications (first or second author):

- **Krishnamurthy, V.**, Cowan, N. B., Fukagawa, M., & Hirano, T. (2026), "Beyond five scale heights: Composition- and cloud-dependent Ariel Tier-2 requirements for sub-Neptunes." arXiv:2609.01577 [*under review*].
- **Krishnamurthy, V.** (2026), "Energetic decoupling from biological scaling: A dimensionless framework for technological civilizations." [*under review in Acta Astronautica*].
- **Krishnamurthy, V.**, Carteret, Y., et al. (2026), "Continuous helium absorption from the leading and trailing tails of WASP-107b." *Nature Astronomy*, 10, 258–270.
- **Krishnamurthy, V.** & Cowan, N. B. (2024), "Helium in Exoplanet Exospheres: Orbital and Stellar Influences." *The Astronomical Journal*, 168, 30.
- **Krishnamurthy, V.**, Hirano, T., et al. (2023), "Absence of extended atmospheres in low-mass star radius-gap planets." *Monthly Notices of the Royal Astronomical Society*, 521, 1210–1220.
- **Krishnamurthy, V.**, Hirano, T., Stefansson, G., et al. (2021), "Non-detection of helium in the upper atmospheres of TRAPPIST-1b, e, and f." *The Astronomical Journal*, 162, 82.
- Hirano, T., **Krishnamurthy, V.**, Gaidos, E., et al. (2020), "Limits on the spin–orbit angle and atmospheric escape for the 22-Myr-old planet AU Mic b." *The Astrophysical Journal Letters*, 899, L13.
- **Krishnamurthy, V.**, et al. (2015), "Computational study of the aerodynamics of the gliding snake *Chrysopelea paradisi*." *International Journal of Applied Engineering Research*, 10, 14476–14479.

Invited summary and research briefing:

- **Krishnamurthy, V.** (2025), "JWST tracks helium escape from the atmosphere of the exoplanet WASP-107b." *Nature Astronomy*, Research Briefing.

Co-authored publications:

- Srivastava, A., et al., including **Krishnamurthy, V.** (2026), "Night-sky emission correction techniques for high-resolution spectroscopy: Demonstration with NIRPS." arXiv:2609.03374 [*under review*].
- L'Heureux, A., et al., including **Krishnamurthy, V.** (2026), "Radial velocity detection of the TRAPPIST-1 planetary system with SPIRou and NIRPS." arXiv:2609.03006 [*under review*].
- Kawauchi, K., et al., including **Krishnamurthy, V.** (2026), "A search for helium in the atmospheres of three sub-Neptunes and a super-Earth around M dwarfs." arXiv:2608.19030 [*under review*].
- Pelletier, S., et al., including **Krishnamurthy, V.** (2026), "The breaking of cold traps and onset of titanium in ultra-hot Jupiter atmospheres: WASP-189b in context." *Astronomy & Astrophysics*, 711, A119.
- Weisserman, D., et al., including **Krishnamurthy, V.** (2026), "Super-Earth masses and stellar abundances from NIRPS reveal tentative evidence for water-rich formation around M dwarfs." *Astronomy & Astrophysics*, 709, A165.
- Wardenier, J. P., et al., including **Krishnamurthy, V.** (2026), "Parametrizing the projected wind fields of ultra-hot Jupiters in thermal emission: an application to GCM spectra of WASP-76b." arXiv:2606.23209 [*under review*].
- Fukuda, I., et al., including **Krishnamurthy, V.** (2026), "The mass of TOI-1883 b: A low-density super-Neptune in the ridge regime transiting an early-M dwarf." *Publications of the Astronomical Society of Japan*, 78, 1602–1614.
- Lamontagne, P., et al., including **Krishnamurthy, V.** (2026), "NIRPS tightens the mass estimate of GJ 3090 b and detects a planet near the stellar rotation period." *Astronomy & Astrophysics*, 706, A278.
- Lacedelli, G., et al., including **Krishnamurthy, V.** (2026), "Sibling sub-Neptunes around sibling M dwarfs: TOI-521 and TOI-912." *Astronomy & Astrophysics*, 705, A260.
- Allart, R., et al., including **Krishnamurthy, V.** (2025), "Complex structure of escaping helium spanning more than half the orbit of the ultra-hot Jupiter WASP-121b." *Nature Communications*, 16, 10822.
- Parc, L., et al., including **Krishnamurthy, V.** (2025), "NIRPS and TESS reveal a peculiar system around the M dwarf TOI-756: A transiting sub-Neptune and a cold eccentric giant." *Astronomy & Astrophysics*, 702, A138.
- Ikuta, K., et al., including **Krishnamurthy, V.** (2025), "The mass of TOI-654b: A short-period sub-Neptune transiting a mid-M dwarf." *Publications of the Astronomical Society of Japan*, 77, 1101.
- Splinter, J., et al., including **Krishnamurthy, V.** (2025), "Precise constraints on the energy budget of WASP-121b from its JWST NIRISS/SOSS phase curve." *The Astronomical Journal*, 170, 323.

- Bazinet, L., et al., including **Krishnamurthy, V.** (2025), "Quantifying thermal water dissociation in the dayside photosphere of WASP-121b using NIRPS." *Astronomy & Astrophysics*, 701, A276.
- da Silva, J. G., et al., including **Krishnamurthy, V.** (2025), "Blind search for activity-sensitive lines in the near-infrared using HARPS and NIRPS observations of Proxima and Gl 581." *Astronomy & Astrophysics*, 700, A177.
- Suárez Mascareño, A., et al., including **Krishnamurthy, V.** (2025), "Diving into the planetary system of Proxima with NIRPS: Breaking the metre-per-second barrier in the infrared." *Astronomy & Astrophysics*, 700, A11.
- Bouchy, F., et al., including **Krishnamurthy, V.** (2025), "NIRPS joining HARPS at the ESO 3.6-m telescope: On-sky performance and science objectives." *Astronomy & Astrophysics*, 700, A10.
- Vaulato, V., et al., including **Krishnamurthy, V.** (2025), "Hydride-ion continuum hides absorption signatures in the NIRPS near-infrared transmission spectrum of the ultra-hot gas giant WASP-189b." *Astronomy & Astrophysics*, 700, A9.
- Allart, R., et al., including **Krishnamurthy, V.** (2025), "NIRPS detection of delayed atmospheric escape from the warm and misaligned Saturn-mass exoplanet WASP-69b." *Astronomy & Astrophysics*, 700, A7.
- Artigau, É., et al., including **Krishnamurthy, V.** (2024), "NIRPS first light and early science: Breaking the 1 $\mathrm{m\,s^{-1}}$ radial-velocity precision barrier at infrared wavelengths." *Ground-based and Airborne Instrumentation for Astronomy X*, Proc. SPIE 13096, 132–148.
- Kuzuhara, M., et al., including **Krishnamurthy, V.** (2024), "Gliese 12b: A temperate Earth-sized planet at 12 pc ideal for atmospheric transmission spectroscopy." *The Astrophysical Journal Letters*, 967, L21.
- Kagetani, T., et al., including **Krishnamurthy, V.** (2023), "The mass of TOI-519b: A close-in giant planet transiting a metal-rich mid-M dwarf." *Publications of the Astronomical Society of Japan*, 75, 529–543.
- Hirano, T., et al., including **Krishnamurthy, V.** (2023), "An Earth-sized planet around an M5 dwarf star at 22 pc." *The Astronomical Journal*, 165, 131.
- Morello, G., et al., including **Krishnamurthy, V.** (2023), "TOI-1442b and TOI-2445b: Two ultra-short-period super-Earths around M dwarfs." *Astronomy & Astrophysics*, 673, A32.
- Delrez, L., et al., including **Krishnamurthy, V.** (2022), "Two temperate super-Earths transiting a nearby late M dwarf." *Astronomy & Astrophysics*, 667, A59.
- Mori, M., et al., including **Krishnamurthy, V.** (2022), "TOI-1696: A nearby M4 dwarf with a $3\,R_\oplus$ planet in the Neptunian desert." *The Astronomical Journal*, 163, 298.
- Kawauchi, K., et al., including **Krishnamurthy, V.** (2022), "Validation and atmospheric exploration of the sub-Neptune TOI-2136b around a nearby M3 dwarf." *Astronomy & Astrophysics*, 666, A4.
- Harakawa, H., et al., including **Krishnamurthy, V.** (2022), "A super-Earth orbiting near the inner edge of the habitable zone around the M4.5 dwarf Ross 508." *Publications of the Astronomical Society of Japan*, 74, 904–922.
- Gaidos, E., et al., including **Krishnamurthy, V.** (2020), "Zodiacal exoplanets in time XI: The orbit and radiation environment of the young M-dwarf-hosted planet K2-25b." *Monthly Notices of the Royal Astronomical Society: Letters*, 498, L119–L124.

Conference Presentations

HWO Japan Science Workshop 2 (presentation) — JAXA, Sagamihara
Atmosphere and escape in the HWO era: What JWST is teaching us — *July 2026*

Exoplanets VI (presentation) — Porto, Portugal
Know thy Star, know thy Planet: Probing TLSE in a M-star — *June-July 2026*

Cool Stars 23 (poster) — Tokyo, Japan
Exploring TLSE in a M-star — *June 2026*

NIRPS Science Meeting (presentation) — Geneva, Switzerland
Young planets with escaping atmospheres — *November 2025*

CRAQ Annual Meeting 2025 (presentation) — Quebec, Canada
NIRISS/SOSS observations of WASP-107b — *May 2025*

Two HoRSEs (poster) — Berlin, Germany
Atmosphere of the young Neptune DS Tuc Ab — *July 2024*

CRAQ Annual Meeting 2024 (flash presentation) — Quebec, Canada
Atmospheres of young planets — *May 2024*

NIRPS Science Meeting (presentation) — Montreal, Canada
Atmospheres of young planets — *April 2024*

NIRPS Science Meeting (presentation) — Montreal, Canada
Search for helium in NIRPS targets — *October 2023*

Exoclimes VI (poster) — Exeter, UK
Near-infrared observations of transiting-planet atmospheres — *June 2023*

CRAQ Annual Meeting 2023 (presentation) — Quebec, Canada
Atmospheres of radius-gap planets — *May 2023*

Japan Geoscience Union Meeting 2022 (presentation) — Online
Atmospheres of low-mass-star planets: Atmospheric escape and beyond — *May 2022*

Exoplanets IV, AASTCS 9 (poster) — Las Vegas, USA
Near-infrared observations of TOI-1235b, GJ 9827b, and GJ 9827d — *May 2022*

53rd Annual Meeting of the Division for Planetary Sciences (iPoster) — Online
High-resolution near-infrared observations of the TRAPPIST-1 planets — *October 2021*

Subaru Users Meeting FY2020 (poster) — Online
Characterization of sub-Neptune and rocky-planet atmospheres with near-infrared spectroscopy — *March 2021*

Rocky Planets in the Era of JWST: Theory and Observation (poster) — NASA GSFC, USA
Exploring the infrared helium line for atmospheric characterization — *November 2019*

Workshop on TESS Follow-up Obs. for Exoplanet (presentation) — Busan, South Korea
New observational results from IRD spectrograph on Subaru Telescope — *October 2019*

Tokyo Planetary Seminar (presentation) — JAXA, Tanegashima
New observational results from IRD spectrograph on Subaru Telescope — *September 2019*

TIEC Research and Presentation 2019–2020 (presentation) — Tokyo, Japan
Ground-based observations of terrestrial exoplanet systems — *April 2019*

TIEC Research and Presentation 2018–2019 (poster) — Tokyo, Japan
Current developments in ground-based exoplanet observations — *November 2018*

AIAA Oklahoma Symposium (presentation) — Edmond, OK, USA
Metamaterial-inspired smart composites for space applications — *April 2018*

6th ELSI International Symposium (poster) — Tokyo, Japan
Concept of a sustainable habitat on Mars — *January 2018*

Master's Thesis Defense (presentation) — Kanpur, India
Modeling the elasto-plastic behavior of aluminum alloys using micro- and damage mechanics — *May 2017*

3rd International Conference on Mechanical Engineering (presentation) — Shanghai, China
MC-DSMC solver for an optimized space-crew re-entry orbital vehicle — *October 2016*

Bachelor's Thesis Defense (presentation) — Coimbatore, India
Monte Carlo simulations for re-entry vehicles — *June 2015*

Invited Talks and Colloquia

Astronomy Colloquium, Tohoku University — Sendai, Japan
Host: Dr. Misato Fukagawa — *July 2026*

Sato Laboratory, Institute of Science Tokyo — Tokyo, Japan
Host: Dr. Bun'ei Sato — *April 2026*

Young Transiting Planet Workshop 2026 — Okinawa, Japan
Host: Dr. Norio Narita — *March 2026*

Astrobiology Center, National Astronomical Observatory of Japan — Tokyo, Japan
Host: Dr. Teruyuki Hirano — *February 2026*

Department of Astronomy & Astrophysics, TIFR — Mumbai, India
Host: Dr. Joe P. Ninan — *December 2025*

Inter-University Centre for Astronomy and Astrophysics — Pune, India
Host: Dr. R. Srianand and Dr. Anupam Bhardwaj — *December 2025*

Department of SPASE, IIT Kanpur — Kanpur, India
Host: Dr. Prashant Pathak — *August 2025*

Department of Earth & Space Sciences, IIST — Thiruvandrum, India
Host: Dr. Anand Narayanan — *July 2025*

Center for Astrophysics | Harvard & Smithsonian — Boston, USA
Host: Dr. Dave Charbonneau — *October 2024*

RIKEN — Saitama, Japan
Host: Dr. Nami Sakai and Dr. Yui Kawashima — *January 2024*

Amrita Vishwa Vidyapeetham — Coimbatore, India
Host: Dr. S. Mahadevan and Dr. Bharat Kishore Sharma — *January 2024*

Indian Institute of Astrophysics — Bengaluru, India
Host: Dr. Sujan Sengupta and Dr. Piyali Chatterjee — *January 2024*

Astrobiology Center, National Astronomical Observatory of Japan — Tokyo, Japan
Host: Dr. Teruyuki Hirano — *May 2022*

Grants and Awards

Canadian Space Agency – JWST Cycle 4 Grant — 2025–present
Project: Probing Stellar Behavior and Planetary Diversity in K2-384
- Canadian Space Agency grant supporting our JWST DDT program in Cycle 4.
- The CAD 60,000 grant supports in the publications and conference participation.

Clark Science Executive Leadership Fund (Clark-SELF) — 2025
McGill University
- Recipient of the Clark-SELF award for executive MBA-level leadership training.
- The CAD 7,000 award supported participation in the Mini-MBA Executive Development Program.

NASA SEEC Grant — 2020–2022
Role: Collaborator; PI: Ravi Kopparapu, NASA Goddard Space Flight Center, USA
- Simulated transmission spectra of atomic helium as a tracer of atmospheric escape.
- The USD 100,000 grant supported the development of helium models.

Monbukagakusho (MEXT) Scholarship for Doctoral Studies — 2018–2021
Ministry of Education, Culture, Sports, Science and Technology, Government of Japan
- Scholarship supporting doctoral studies in exoplanet atmospheres at the Tokyo Tech, Japan.
- The scholarship provided JPY 6,090,000 over 3.5 years.

GATE Stipend for Master's Studies — 2015–2017
All India Council for Technical Education scholarship, Government of India
- Scholarship supporting master's studies in aerospace engineering at the IIT Kanpur, India.
- The scholarship provided INR 300,000 over two years.

Current and Past Projects

Atmospheric and Orbital Architecture of Close-in Packed Planetary Systems — 2023–present
Role: Principal Investigator
Investigating the atmospheres and orbital architectures of compact, close-in multiplanet systems. Systems such as TRAPPIST-1 and K2-384, which contain several planets in or near orbital resonances, provide valuable insights into planetary formation and evolutionary pathways.

NIRPS – Search for Blue Planets Around Red Stars — 2022–present
Role: Instrument and Guaranteed Time Observations team member
Member of the NIRPS instrument team and observer since 2022. The project focuses on transit observations of young planets orbiting red dwarfs using the Near Infra-Red Planet Searcher on the

ESO 3.6-m telescope at La Silla Observatory, Chile.

Subaru Strategic Program – InfraRed Doppler Spectrograph 2019–2024
Role: Co-investigator; PI: Bun'ei Sato
Member of the Subaru Strategic Program team since 2019. The program focuses on discovering Earth-mass planets in the habitable zones of mid-to-late M dwarfs and determining the distribution of planets around cool stars. The program comprises 175 observing nights over five years with the high-resolution IRD spectrograph on the 8.2-m Subaru Telescope. My work focuses on stellar characterization using the He I triplet at 1083 nm.

Simulating Transit Spectra of Atomic Helium as a Tracer of Escape 2020–2022
Role: Collaborator; PI: Ravi Kopparapu
The project updated a 3D upper-atmosphere model to calculate atmospheric escape rates and predict transmission spectra for transiting exoplanets around a range of stellar types. The model incorporated atomic absorption lines, particularly the metastable helium triplet at 10833 Å, which traces the upper-atmospheric region from approximately μbar to nbar where escape is launched.

IRD General Observing Programs 2019–2022
Role: Principal Investigator and Co-investigator
Secured more than 20 hours of Subaru Telescope time as Principal Investigator, in addition to several observing nights as a Co-investigator, using the IRD spectrograph to investigate photoevaporation in sub-Neptunes. The programs resulted in more than three peer-reviewed publications.

Research Supervision

Graduate Supervision, IIST 2025–2026
Co-supervisor for Arya Raj with Prof. Anand Narayanan and Prof. Mathew
Project: Feasibility of detecting oxygen with the Thirty Meter Telescope
- Simulated the atmosphere of a habitable-zone Earth-like planet orbiting a Sun-like star.
- Assessed oxygen detectability using high-resolution cross-correlation spectroscopy.

Graduate Supervision, Université de Montréal 2024–2026
Co-supervisor for Mathis Bouffard with Prof. René Doyon and Prof. Nicolas Cowan
Project: High-resolution transit spectroscopy of the TRAPPIST-1 planets
- Expanded Mathis's undergraduate research to include the complete TRAPPIST-1 planetary system.
- Analyzed transit observations obtained with CFHT/SPIRou and Subaru/IRD.

Undergraduate Supervision, University of British Columbia 2025–2026
Co-supervisor for Sean Collins with Prof. Michelle Kunimoto
Project: Confirmation of a TESS planet candidate
- Analyzed TESS observations and ground-based photometry to confirm a TESS Object of Interest.
- Supported the preparation of radial-velocity observations with a high-resolution spectrograph.

Undergraduate Supervision, Amrita Vishwa Vidyapeetham 2024–2026
Supervisor for Anusha and Ashfaq Ali (non-thesis research)
Project: Transit-timing variation analysis of a multiplanet M-dwarf system
- Investigated a multiplanet system around a M dwarf to constrain planetary masses using TTVs.
- Combined Kepler, TESS, and ground-based photometry to measure the transit-timing variations.

Undergraduate Honors Research Thesis, McGill University 2023–2024
Co-supervisor for Mathis Bouffard and Maddy Walkington with Prof. Nicolas Cowan
Project: High-resolution transit spectroscopy of TRAPPIST-1b
- Co-supervised the analysis of CFHT/SPIRou transit spectroscopy of TRAPPIST-1b.
- The research was presented at the Emerging Researchers in Exoplanet Science Symposium IX at Cornell University, Ithaca, New York, in July 2024.

Technical and Professional Skills

Software and GitHub Packages GitHub: VigneshAstro

HELIX – Helium Exoplanet Literature Investigation Xtractor 2024
Python wrapper for extracting helium-related papers from arXiv
- Developed HELIX, a Python wrapper for efficiently extracting research papers from arXiv.
- Created an accompanying Jupyter notebook with instructions for customizing search terms and extracting literature related to other atmospheric species, including hydrogen and water.

Signal-to-Noise Ratio Simulator for NIRPS and HARPS 2023

Exposure-time and signal-to-noise simulator for HARPS and NIRPS

- Developed a signal-to-noise ratio simulator for the HARPS and NIRPS spectrographs on the ESO 3.6-m telescope at La Silla Observatory, Chile.
- Improved the ETC and implemented planetary velocity-smearing models for transit observations.

Observational Experience

Support Astronomer: ESO 3.6-m Telescope 2022–present

Instruments: NIRPS and HARPS

- Member of the NIRPS instrument and Guaranteed Time Observations team, focusing on the atmospheric characterization of transiting exoplanets.
- Provided on-site observing support at ESO's La Silla Observatory and participated in remote observations for the NIRPS GTO program.

Support Astronomer: Subaru Telescope 2020–2022

Instrument: InfraRed Doppler spectrograph

- Worked as a support astronomer for the IRD spectrograph on the Subaru Telescope at Hawaii.
- Served as a remote observer for the Subaru Strategic Program from the NAOJ Office in Mitaka.

Peer Review

Reviewer for Peer-Reviewed Journals 2021–present

- American Astronomical Society journals: *The AJ*, *The ApJ*, and *The ApJL*.
- *Monthly Notices of the Royal Astronomical Society*.
- Nature Portfolio journals, including *Nature* and *Nature Astronomy*.

Professional Service and Committees

Exoclimes VII and ExoSlam 2025

Local Organizing Committee and Summer School Organizer; Montreal, Canada

- Served on the Local Organizing Committee for the seventh edition of the Exoclimes conference.
- Contributed to the organization and delivery of ExoSlam, a summer school designed for graduate students and early-career researchers.

Gemini Strategic Science Plan Working Group 2024

Exoplanet Representative; online

- Appointed to the working group responsible for drafting recommendations for the Gemini Board of Directors and shaping the observatory's strategic plan for the 2030s.
- Represented the priorities and perspectives of the exoplanet and Canadian astronomical communities.

Two HoRSEs July 2024

Local Organizing Committee; Berlin, Germany

- Served on the Local Organizing Committee for the second edition of *High-Resolution Exoplanet and Stellar Characterization Today and in the ELT Era*.
- Supported the Scientific Organizing Committee with conference operations, participant coordination, and logistics.

Trottier Space Institute Seminar Coordinator 2023–2024

Trottier Space Institute, McGill University; Montreal, Canada

- Coordinated institute seminars and hosted visiting researchers working across astronomy, astrophysics, and cosmology.

Media and Interviews

A complete archive of interviews and media coverage is available here.

WASP-107b – Extended Helium Tail 2025

- BBC Tamil: Extensive helium tail around WASP-107b and interview with Dr. Krishnamurthy.
- McGill University: Exoplanet observed shedding its atmosphere in real time.
- Trottier Institute for Research on Exoplanets: Exoplanet shedding its atmosphere in real time.
- Science Springs: Exoplanet observed shedding its atmosphere in real time.
- Space.com: A "super-puff" exoplanet is losing its atmosphere, and JWST had a look.
- MyScience: Exoplanet observed shedding its atmosphere in real time.
- Time.news: Helium leaks from exoplanet WASP-107b.
- Mirage News: Exoplanet spotted shedding its atmosphere in real time.
- ScienMag: Helium absorption detected in WASP-107b's tails.

Gliese 12b – Discovery July–August 2024

- The Conversation: The discovery of a new Earth-like planet could shed further light on what makes a planet habitable.
- Puthiya Thalaimurai: Discovery of a potentially habitable-zone exoplanet.
- Puthiya Thalaimurai: Video report on the discovery of Gliese 12b.
- Trottier Institute for Research on Exoplanets: Discovery of an Earth-sized exoplanet in the habitable zone: Interview with Dr. Krishnamurthy.
- McGill University: Discovery of a new Earth-like planet could shed light on what makes a planet habitable.

New Star Formation Observed with JWST November 2023

Contributor; Puthiya Thalaimurai; author: Jayashree; language: Tamil

- Contributed scientific context for an article about the formation of the protostar HH 212 as observed with JWST.

TRAPPIST-1 in the Modern Era of Astronomy November 2023

Contributor; Puthiya Thalaimurai; author: Jayashree; language: Tamil

- Contributed to a review of the TRAPPIST-1 system and current knowledge of its planetary architecture and atmospheres.

What Is a Solar Storm? August 2023

Contributor; Puthiya Thalaimurai; author: Jayashree; language: Tamil

- Explained solar energy production, the physical origin of solar storms, and the role of ISRO's Aditya-L1 spacecraft in monitoring the Sun.

Additional Professional Experience and Outreach

Market Research Analyst 2024–2025

GLiN Corporation, Japan

- Conducted market research for projects involving India and North America.
- Collected and analyzed data and prepared recommendations based on market trends and client requirements.

Public Science Outreach 2023–2025

Trottier Space Institute, Montreal, Canada

- Contributed to public astronomy and science outreach activities as a member of the Trottier Space Institute outreach team.

SHAD Canada Recruiter September 2023–2025

Montreal, Canada

- Promoted experiential education and youth development programs to high-school students and encouraged participation in SHAD Canada's summer programs.

Part-Time Science Teacher March–September 2021

RED Program, Musashi High School, Tokyo, Japan

- Taught a range of science subjects in English to Japanese high-school students.

Data Analyst Intern Dec 2020–June 2021

Data Visualization Laboratory, Tokyo, Japan

- Developed visual representations of astronomical data for public education and science-outreach projects.

Part-Time English Teacher Nov2020–Sept 2021

Tokyo Global Gateway, Tokyo, Japan

- Taught more than 500 elementary-, middle-, and high-school students from Tokyo and the surrounding region.

Teaching Assistant July–September 2020

Tokyo Institute of Technology, Tokyo, Japan Instructors: Prof. Kumara, Prof. Stacey Doremus, Prof. Eri Ota, and Prof. Rie Murakami

- Supported the collaborative Georgia Tech–Tokyo Tech problem-based learning courses.

Content Creator – edX 2019–2021

Center for Innovative Teaching and Learning, Tokyo Institute of Technology Supervisors: Prof. Jeffrey S. Cross and Prof. Nagahama

- Served as a Graduate Student Assistant–Program Developer and contributed to the development of online courses for edX.

President Aug 2019–Aug 2020

Indian Students' Association of Tokyo Tech.

Distribution Manager Sept 2017–Sept 2019

Habitable Press, a publishing imprint focused on Earth, space, and the future.

Research Intern May 2017–April 2018

Blue Marble Space Institute of Science, USA.

Teaching Assistant January 2016–June 2017

Indian Institute of Technology Kanpur, India

- Conducted tutorial sessions and supported instruction in Engineering Graphics and Aerospace Structural Analysis I.

President – Astronomy Club June 2013–May 2014

Amrita Vishwa Vidyapeetham, India

- Organized and conducted night-sky observing sessions near Coimbatore, India.

Science Outreach Program January 2012–May 2015

- Encouraged students from multiple districts across Tamil Nadu to pursue education and careers in science.